Victor Haghani on Death of Random Walk, and Passive, Buybacks, and LTCM

Podcast: Monetary matters

AI Executive Summary

This transcript features a deep dive into market mechanics, investor psychology, and asset allocation strategies between an interviewer (Jack) and Victor Higginbotham of Elm Wealth.

Below is a structured summary of the key intellectual arguments presented in the conversation.

---

1. The Macro Impact of Stock Buybacks

The discussion explores whether share buybacks only benefit the specific company performing the buyback or if they lift the entire market. The argument suggests a two-fold impact:

- **The Index Fund Mechanism (Passive):** When a company buys back shares, index funds—which must maintain specific weightings—sell those shares. This creates a surplus of cash within the fund, which is then redistributed to buy other companies in the index. Theoretically, this could boost the prices of all *other* stocks while leaving the buyback company's price unchanged.
- **The Active Investor Mechanism (Active):** For "stock pickers," a buyback creates upward pressure on the specific stock's price because shareholders may refuse to sell due to capital gains taxes or fundamental belief in the company. When these investors eventually do sell, that capital flows into other equities.
- **Conclusion:** The net effect of widespread buybacks is an upward lift on the entire market.

2. "Extrapolators" vs. "Momentum Investors"

One of the most complex parts of the discussion is the distinction between two types of trend-following behaviors. While they sound similar, their impact on market stability differs significantly:

- **Momentum Investors (The "Binary" Strategy):** These investors use a discrete, "chunky" trading rule. For example: *"If momentum is positive, I move from 50% to 75% exposure; if negative, I drop to 25%."* They are essentially "price takers" who react to a specific signal. Because their moves are discrete and they don't necessarily drive the trend through continuous buying, they can often perform well by front-running the predictable impact of others.
- **Extrapolators / Return Chasers (The "Gradient" Strategy):** These investors estimate future returns based on a sliding scale of past returns. As markets improve, they incrementally increase exposure (50% $\to$ 55% $\ast \dots \to$ 75%). This continuous, reinforcing behavior creates significant price impact and drives the self-reinforcing cycles seen in bull and bear markets.
- **The Performance Gap:** The speakers suggest that momentum strategies tend to do well because they benefit from the predictable price movements caused by extrapolators, whereas return chasing (extrapolating) tends to underperform because the very act of chasing returns drives prices to levels that are unsustainable.

3. The "Five-Star Screen" for Asset Selection

Victor describes a rigorous framework used at Elm Wealth to decide which risky assets are worth owning. To be included, an asset must meet five criteria:

1. **Systematic Risk Premium:** There must be a logical reason why investors deserve a premium (i.e., the asset carries "nasty," non-diversifiable risk).
2. **Verifiability:** The expected risk premium must be estimable. This excludes commodities like oil, which lack predictable cash flows.
3. **Pure Systematic Risk:** The investor seeks broad exposure rather than idiosyncratic (company-specific) risk. This moves the preference away from single stocks (like NVIDIA) toward broad indices.
4. **Liquidity and Transparency:** The asset must be easy to trade and easy to understand.
5. **Efficiency:** The asset must be low-cost and tax-efficient.

4. Elm Wealth’s Current Market Outlook

Based on their dynamic asset allocation model, the speakers outline their current positioning:

- **US Equities (Underweight):** They are underweight US stocks because the long-term expected return relative to safe assets is currently very low (approx. 1%), and valuations are high compared to historical earnings trends.
- **Non-US Equities (Overweight):** They prefer international equities due to higher earnings yields (better valuation) and positive momentum.
- **Fixed Income:** They are primarily in Treasury bills, remaining underweight in nominal bonds and TIPS because those assets are currently experiencing negative momentum/volatility.
- **The Strategy Core:** The strategy is not "predicting" the future via complex forecasting, but rather using a transparent, rules-based approach to adjust exposure based on long-term expected risk premiums and momentum as a proxy for market risk.

Full Transcript

So I feel like we could talk about LTCM. We could talk about, I mean, let's not do that. That's so boring. I mean, everybody's sick and sick of LTCM. Yeah. Yeah. So what do you think the real lessons are? You know, I think the biggest lessons are about personal risk taking. The first big lesson has to do with skin in the game. Maybe the next lesson is that running a leverage pool of standalone capital might be a bad business structure. You know, it might be better to do relative value trading. within larger institutions where it's a small part of the activity and where it's not exposed to financing risk, etc., from all the counterparties that are funding that activity. So I think those are two things. I think relative value trades tend to have fatter tails than what we call delta one trades. That's kind of natural.

Something that we knew about but I think that was more highlighted by that. I don't know I think those are like three really really big Important lessons from from that. I think the main I think probably the main Difference is that they run tight stop losses on all of their strategies And so they are trying to force all of their strategies to be liquid enough that they can operate tight stops I think that's really helped them. I think that that helps bring a positive exposure to momentum into their strategies. And I think that's probably the biggest divergence between what they're doing and how they've been more successful navigating a number of crises than what we were doing back then. We didn't have tight stop losses on our positions. Felt like relative value trading, it's hard to put tight stop losses because they tend to be kind of illiquid and they tend to look more attractive as they're starting to widen out.

But yeah, and I think that probably the pod shops have a greater, a broader mix of strategies than we were running at LTCM. And that's also helped. But, you know, I mean, some of the pod shops that were some of the hedge funds that are have done successfully had really big drawdowns, like existential drawdowns in 2008, 2009, and just managed to survive them, which is great. but that some of them were vulnerable. And at other times, some of the basis trades have looked a little bit dicey and put some hedge funds in a precarious position, but they all worked out. And again, LTCM

probably could have worked out if LTCM had not been so much the focus of the financial markets at the time. Like if LTCM had sort of been more quiet, had been quietly on the side and people didn't know that we had had such a big drawdown and that we were likely to be unwinding a lot of positions. Probably LTCM would have survived as many other relative value hedge funds survived, though with large losses in 1998 as well. Join today by Victor Hagani of Elm Wealth. He's the author of The Missing Billionaires, the author of an upcoming book. He'll tell us about that as well as he was the founding partner of Long Term Capital Management. Victor, welcome to Monetary Matters.

Thanks very much, Jack. Co- co-author, by the way, with my partner James White on that book and the forthcoming one, too. Tell us about who killed the random walk. What is the random walk? What is the work that you've been doing that shines new light on it? And why does it matter? Oh, thanks for asking that, Jack. This is something I haven't I don't think I've talked on any. really publicly about this except we've been talking at a number of seminars about this paper. It's a piece of research that we've been working on for three or four years and it just got accepted into the Journal of Investment Management. It's available as a working draft on SSRN. And basically the starting point is that everybody who follows the stock market sees a lot of behavior in the stock market that is hard to reconcile with the idea of everybody is a rational, fully informed agent making long-term investment decisions based on expected cash flows of the stock market, which is the classical financial economics description of the stock market and asset pricing theory to begin with at least. And on the other hand, we have behavioral economics that came up relatively recently and behavioral economics Sheds a lot of light on the ways that we make the people make weird decisions, but it tends in its extremist form. It doesn't really Give us much predictions or things that we can test in terms of market behavior So what are these puzzles? I should say of start there. What are the different puzzles of the stock market? Well, I think the biggest puzzle of all is that why is the stock market so much more volatile than the volatility of long-term earnings. This is the famous Schiller Campbell work from the late 80s, partly responsible for Schiller's Nobel Prize, where he pointed out that the stock market is so much more volatile, is twice as volatile as expected earnings are. A little bit of a tricky thing to exactly measure, but I think that

he did that pretty convincingly. And I think people feel that way. And that's become accepted within Practitioners and academics alike as as being the way things are that stock market so much more volatile then some people say well That's just because the discount rate is also moving not only are people's expectations of earnings changing But the discount rate is too but that doesn't really answer the puzzle at all. It's just reframing the puzzle Other puzzles that people think about with the stock market is why is volatility so so volatile itself Why do we go through periods of extreme volatility?

And then it kind of stays that way for a while. It's clustered. And then we go to periods of peacefulness. Why do we have trends? Why has momentum investing been such a good way to invest in the stock market? Why has value investing not been so great, been so great when the PE is high? If you reduce your exposure to stocks, yeah, it tends to be okay sometimes, but a lot of times the PE is high and stock market just keeps on going up. Why do we get booms and busts? Maybe that should have been my first one. Why do we get booms and busts, this exuberance and kind of depression and stocks at different times? And several more. Well, our paper, which I should say really builds upon a lot of previous research that's been done. Our paper is titled, Who Killed the Random Walk? Why Extrapolators Explain booms and busts and other stock market puzzles and anomalies. And our research in this field is basically was developing what's called a multi-agent or heterogeneous belief model where different investor types Come up with their belief about stock market returns differently So you'll have one group of investors will call them value investors and they're really thinking about the long-term earning streams of equities and dividends and they want to own more equities when the expected risk premium based on these long-term cash flows is higher and vice versa when the expected return is lower. Other investors, which is probably the predominant investor class these days, are static investors, which they're just like investing in a target date fund or a balanced fund. It's going to be 60-40 more or less all year long, and they're not going to change. They just want to stay at 60-40 and rebalance there every month, every quarter, whatever.

And static asset allocators are a huge component of the market these days. It's not just 401k and IRA investors in these balance and target date funds.

It's a lot of institutions. The Norwegian oil fund is pretty much a static asset allocation investor. And these kinds of investors have not been modeled very much in this multi-agent body of literature. Finally, I mean, well, we've modeled a bunch of other investor types, too. But the third type, which is really the key one here, are extrapolators or return chasers. And what do they do? They basically come up with their expected return for stocks based on recent history, based on returns over the last five years is how we've modeled it with kind of an exponential decay.

And so when returns have been good in the stock market, extrapolators get more bullish about returns in the future. We dampen it and try to be realistic about it if stocks went up 25% the last year, that people aren't saying, oh, I expect 25% next year. But they're more bullish. They expect a higher expected return, a higher risk premium, and they want to own more stocks. And there's a ton of Survey data etc that shows that many investors do behave this way and and and anecdotally I think we all see it and and and and depending on the marketplace is stronger or weaker like when it comes to investing in cryptocurrencies people are very tied to history when it comes to investing in the two-year note There's not too much extrapolation going on so like I go ahead go ahead Well, anyway, so we take these investor types. We put them together into a hypothetical simulation. We let them trade with each other to do to get to where they want to be. We introduced shocks to earnings. We put in some noise, some noise in that sometimes people need to sell to sell some equities to buy a house. Other times people are getting compensation and need to invest in equities. We try to create a realistic environment where these people can all trade with each other and and voila what we get at the end is like all is an explanation of like all these different puzzles that this dynamic system gives us excess volatility it gives us stochastic volatility it gives us trends it gives us booms and busts all these different things and it does it with like a reasonable set of parameters there's a lot of parameters in this model so you could say well geez Once you have this many parameters, you can explain anything. And that's true. But it still is kind of comforting as a framework for thinking about things that this is probably a reasonable description. And you can ask a bunch of different questions like how inelastic is demand as a function of price? Like if somebody comes in and needs to sell 100 billion of equities, How much is that likely to move the price of the stock market? Because

these three investor types have to adjust their holdings to absorb this 100 billion of supply or demand that comes to the market. So it's a really rich framework for asking and answering potentially a lot of questions. And yeah, we've found it super exciting and it's been gotten a good reception. I think we've presented the paper about five different seminars and we're super excited that the Journal of Investment Management is going to publish it. So many things I want to get into, Victor, just starting with the title. So the random walk theory states that asset price change is completely random and unpredictable, meaning past market movements cannot be used to forecast future prices. Are you challenging or are you accepting that theory in the paper to be clear?

Well, I'd say we're challenging it. I think that the stock market really doesn't feel like the classical random walk that doesn't doesn't feel like the classical random walk that we all learned about in in our finance courses years ago. So yeah, we're challenging that the stock market follows a random walk and a core of that I suppose would be momentum like Jim Simons RIP that had phenomenal returns like best returns ever in their core fund at Renaissance of it was it it was Very complicated, but essentially it was a it was a momentum strategy, right? And like a lot of a lot of incredible returns hedge fund returns from a while ago are From momentum up like if it was up Monday Tuesday Wednesday It's a decent bet to go bet that it's going up on Thursday. It's a lot more complicated with that But like momentum is a giant thing and that that is one of the things that kind of disproves the random walk. And you in your paper, you have like eight other puzzles, other than momentum, I think seven other puzzles, and you're trying to explain them. Yeah. And I think you hit on the one that probably is the biggest challenge to the random walk. And it's the one Jean Fama, the father of the efficient markets hypothesis, right? Or one of the fathers.

Gene Fama says momentum is the mother of all anomalies, but it is an anomaly. It is weird that we see it and we see it. Researchers, a number of researchers have shown that like momentum exists in everything. It's in natural gas. It's in stock prices. It's in 30 year long bond prices. It's in Bitcoin, but it's everywhere. It's been everywhere. It continues to be there. And what the heck? How can that? How can momentum be there? And then, as we were talking about earlier, that stop losses, if you follow a stop

loss program, investors that are following a tight stop loss program are kind of like, I don't want to say they're closet momentum investors, but they are exposed positively to momentum and it helps. So when you go back and think about the turtles, remember the turtles from the commodity? It was kind of like the story from trading places.

where these commodity traders, successful commodity traders were like, I bet we could just get some kids out of high school or college and teach them to be successful commodity traders. And they hired all these people that knew nothing about anything. And they got them to be really successful by forcing them to just follow a tight stop loss. They could do whatever they want, but they had to cut their losses and let their profits run. And it was almost like whatever they did, as long as they did that, they made money. And that's because those markets had momentum, exhibited momentum over that period. So why is it that momentum works? All the quantitative geniuses from 40 years ago, they said, okay, let's do momentum. Why hasn't this been kind of competed out of the market in the same way that, okay, value was a thing. You could make money in 1970 by buying giga cheap stocks. And now that strategy works a lot less to put it mildly. Why isn't that true for momentum? Why does momentum still work?

it's a question of relative capital sizes. So if the momentum we believe is created by the behavior of extrapolators, if the extrapolators have much, much more capital at their disposal, if they're having much more impact than the pool of capital that's doing momentum, then the momentum investors won't arbitrage or compete away this momentum effect. And you have to realize that Momentum is a really interesting investment strategy because on the one hand, it has really good returns and people love that. On the other hand, it just seems like the dumbest thing you could ever do. What? The market went up. I'm going to buy more. I'm going to buy. The market went up. I'm going to buy. Like, isn't the whole idea of investing to buy low and sell high? Well, what's momentum? Momentum is buy high and sell higher. It's sell low and sell lower.

And it just is this thing that just seems so strange to investors that it impedes or limits how much capital is dedicated to it, because it just seems

like such a weird investment strategy to most people. And as a result, when momentum goes through periods of time, when it's not doing well, everybody is like, I knew that was a stupid strategy. Why did I ever invest in that to begin with? And people just pull their money out of it so fast. because it's going through some bad period of time. So I don't know. I mean, I think we could get to a point in time when everybody is thinking about the market through this lens of this heterogeneous belief model with return chasers and so on. And there's just so much capital that starts to follow this more binary momentum strategy that momentum goes away forever. But as I say, it's a question of capital. And I don't know. I mean, I think.

And in our model, if we give the momentum investors in our model more capital than the return chasers, than the extrapolators, well, then the momentum guys don't do well. And that's that. So you can see it immediately within our little toy model that once momentum guys get too big. If momentum guys are just price takers, the world is happy for them. When they start to really have a lot of price impact, the world is not happy for them. Yeah. And I think also on a fundamental basis, momentum is also a thing. Like a company that grows its earnings at 20% and there's an acceleration is more likely to do that than a company that their earnings are doing really badly. Yeah. Yeah, I think there's also momentum in fundamental economic variables, I think is what you're saying, right? Yes. Yeah, yeah.

Yes, okay, so I mean in your kind of when you laid out there's so many interesting things, but I wrote Something down on it's funny. I like I don't have a notebook in front of my notepad where he has this Chinese I don't even know what it is probably my business partner, but So about you said the word static Investors that to me that means index investors passive investors Tell me about the significance of those of the rise of passive investors who really just buy the index, the S&P, the MSCI world, whatever index you want, how does that change the investment universe and be as provocative as you can? Because I've got a lot of potential pushbacks and challenges. And it's emotional to me. So we'll get into it. Let's have a fight, Victor. I want to have an intellectual cage match with you. Okay, let's do it. Okay. So I think that it's really important to distinguish between two kinds

of potentially passive investing. One of them is, I'm going to invest in the stock market and rather than giving my money to somebody that decides which stocks to buy and owns a subset of stocks, I'm going to give my money to a market cap weighted index fund. So that's index investing as a means of stock picking, let's say. In other words, it's not stock picking, it's just investing in all the stocks when I decide to invest in that. The other kind of passive investing is passive asset allocation and I think that passive asset allocation is So I'll just I'll just make my statement passive asset allocation is really problematic It doesn't make sense passive asset allocation. It doesn't make sense Modern portfolio theory never said that anybody should be a passive asset allocator the whole capital asset pricing model was had this insight that people are going to be wherever they want to be based on their preferences on that capital asset pricing line. They're not going to be passive. They're going to make an active decision based on the expected return risk and their risk preferences. So passive. I'm just going to jump in and explain. Sorry. So passive for stocks means I'm not picking stocks. I just buy the S&P 500 or some other index. Passive asset allocation for overall assets is I'm not going to choose my stock allocation, my bond allocation, my hedge fund, private equity commodity allocation. I'm just going to do what some index told me. So I'm going to be 60% stocks, 40% bonds, or if you're an endowment, like 30% stocks, 10% hedge funds, 20% blah, blah, blah, blah, blah, blah, blah, blah. Yeah, passive asset allocation means I'm going to just choose some allocation of stocks and bonds.

And I'm just gonna stick with that, despite the fact that I can see that everything is changing all the time, that real interest rates are changing, that the earnings yield, the long-term expected return of stocks are changing. If stocks are super cheap and the S&P has a PE of five, so stocks are super cheap and bonds are super expensive, I'll be 60-40. If stocks are super expensive and bonds are super cheap, Also be 6040 you're not making any changes you say that so I would call that static asset allocation But for the purpose of this conversation, let's call that passive asset allocation. I think it makes no sense I think it's I think that's responsible for all the things that people are worried about With with respect to the word passive. I think it's problematic on the other hand I think that investing in the stock market using market cap weighted index funds is mistakenly criticized for creating

problems. I think that creates very few problems. I think it's a good thing for most investors. If you have some evidence or belief that you can pick stocks, go for it. But I think that investing in index funds is a way to get stock exposure is totally fine. And I think the criticisms of it are mostly misguided, are mostly coming from people that have had their businesses hurt by that.

because money has been taken away from them. And so that's as provocative as I'm going to be for now. I think that index funds as a way to invest in the stock market is fine. I think that passive asset allocation, where you're going to just decide to be 60, 40, 70, 30, whatever, is really a problem for our markets. And it's giving us these booms and busts. At least it's giving us a boom. It hasn't given us the bust yet. And I think it's problematic. And I think those are two really different things. And I think that critics of indexing are missing the mark because they're saying, oh, it's that when somebody invests in an index fund, look how much NVIDIA they're buying. This is a problem. No, I don't think that's the problem. I think the problem is passive asset allocation, not passive stock picking or no stock picking. OK. So I feel like on the passive stock point, I agree with you. I think that a lot of critics of passive investing, they say by being a fundamental bottoms up investor who focuses on value, who focuses on quality. When I was growing up with the business in 1994, in 2001, that worked before the markets have been perverted by the rise of passive investing. And now what I used to do good investing doesn't work anymore.

I struggle with that because to me, I look at the fundamental earnings power and the companies whose earnings go up the most, generally their stock prices have gone up the most relative to the expectations that were baked into. Like to me, the markets don't seem wildly inefficient in terms of like, oh, the market is, we're all, the market is. believing in the lies. Like there's all this been price distorting aspects. Like I, if I look at the biggest stocks in the market, like their earnings have grown tremendously and they're the biggest in the market, like with the exception of Tesla. Now, maybe you could say, okay, Tesla stock has been flat over the past three years while it's earnings has gone down, like in a proper market, like the valuation should have gone down and the price should have gone down even more because

earnings went down. But I don't really see like, to me, there are distortions in the market.

that are caused by 10 things and those by 10 forces that are greater than passive, like, like for example, stock promotion. An IPO comes out. So everyone who wants to unload their bags on everyone talks incredibly bullishly on it. The human nature to miss to, to, to be overly optimistic on the upside and overly pessimistic on the downside. To me, those are the root causes of inefficiency in the market. not passive investing. So I'm disappointed that I'm agreeing with you as much as I am. I want to have an intellectual cage match. I'm so happy that we're agreeing. I mean, I'm so, so happy that we're agreeing on this. Yeah. I mean, what you just said, I totally agree with it. 100%. It's like the problem isn't that when it comes to the cross section of stocks, when it comes to the valuation of individual stocks, I think the problem is not indexing. Or if it is, it's number 11.

on on the list like okay maybe oh whatever like uh the relative liquidity of some big stock is less than the relative liquidity of some other stock whatever fine but maybe but those 10 things there's 10 things exactly as you said i mean it's just all this individual craziness like oh i'm an all-in tesla investor i mean remember that guy that was in the wall street journal i think his name was like uh mr volt or something That was his last name was like Volt or Mr. Watts like wattage. And he was an all in Tesla investor because his name had to do with electricity. Well, yeah, I mean, if money comes into the market and it's just buying the market cap index, who's selling to them? I mean, maybe it's just people that own the market cap index on average are selling. And remember that all active investors combined.

have a portfolio that is the market portfolio. So if an index fund buys and it's just buying from a random selection of active investors, if it's buying from an index fund that has no impact, if it's buying from some random selection of investors out there, well, they own the market portfolio. So that has no price impact either. So yeah, I think, but I think that I thought maybe your question was even going in a different direction, which I had never thought of before, which is like, Why does the market more or less, for most stocks, more or less for most stocks, why do most stocks look reasonably priced in this world where there is so much craziness going on,

where we have zero commissions and tremendous volume and all of this? And I think that I hadn't really considered this question before, but I think the answer to that is that the hype and the excitement and everything of retail investors is focused on just a very small fraction of names. So we have a small fraction of names that probably are misvalued by fundamentals, but 95% of stocks are not, like they don't care about them. They don't care about, I don't know, but they don't care about Pfizer, maybe, I don't know. They don't care about whatever. Like they care about, they care about Sandisk. They care about, they care about.

IBM energy, they care about Standis, they care about Micron, they care about Tesla, they care about SpaceX. They don't care about Hilton. They don't care about Hilton and other investors whose names we do not know care about Hilton. Yeah, no one else. And so those prices are kind of being set by fundamental investors more or less by stock pickers and by sensible stock pickers that are trying to figure things out. And so I think in the bulk of the market, that the prices kind of look kind of reasonable. And then around the edges, there's some craziness, but in aggregate it kind of looks okay-ish. But it's the overall level of the market that I think looks kind of ridiculous right now. I mean, that when you look at the capital market assumptions of 20 investment management, bank, investment bank, brokerage, Vanguard, BlackRock, Goldman, everybody, right?

They're saying that the expected long-term return for US equities is like 6%. Well, holy cow, 30-year treasuries are around 5%. What? That's just 1% more. That's our risk premium today. And this is the consensus among a bunch of people that have businesses that benefit from being bullish on stocks. So I think you almost have to pay. Yes, but the real fees are in hedge funds. particularly private equity and private credit. So if you say, oh, stocks are only going to do 2% a year, then you get your clients into the private equity, private credit. And that's where the real fuse are. That's my conspiracy theory. I don't know. Well, except for the fact that that's not Vanguard's gig. That's not Black Rocks gig. That's not Bank of America's gig is a little bit Black Rocks gig, but the other ones I agree. Yes. Yeah. But I mean, it's just very minor. I think, and I think this is like either, if anything, Okay, maybe that offsets this bullish tint that I would argue. So maybe they try to say returns are going to be worse to get people into other

things. But on the other hand, they also have businesses that benefit from people being excited about stocks. So whatever. So the blue chip consensus, long-term forecast for stocks from this level are 6% to be clear. Are you saying you agree with that or that's too high or too low and why?

I agree that that's the appropriate expectation based on where we are. And I think that's like voluntary financial repression. And I think it's like, I mean, I think that's a shame. It really is going to make it hard for people to grow their wealth if stocks are just giving 1% more than safe assets. So I think it's unfortunate. I think it's realistic. I think it's, and I think that it's being caused by All of this static asset allocation out there that just people are like, I'm just going to be 60 40. It served me well. I don't care that I'm just going to get 1% more. And so we have Berkshire Hathaway is like, all right, we're going to reduce our stock exposure guys, but nobody, but very few other investors are willing to do that. And anybody who has done that kind of has mud on their face. And so those, those people is like the jet is the widowmaker trade they call it from.

That's what they used to call the Japanese interest rate trade where so many people were shorting Japanese interest rates at 1% and eventually there was nobody left because everybody lost their jobs who did it then so Yeah, I just think it's realistic, but but but unfortunately low and meanwhile non-us equities at least are offering better long-term returns again our belief and the consensus belief is that non-us equities offer healthier long-term returns relative to safe assets, but But here we are with very elevated US stock prices. And why? Why do we have that? Well, I don't know if you were going to ask me why. I would say why is because of we have so many extrapolators. And these extrapolators are just more bullish than ever on stocks. We have more leverage through margin low. You can see retail investors are so bullish on stocks. This is typical after a period of stocks doing well.

And meanwhile, up until recently, what have we had? We've had companies buying back over a trillion dollars of their stocks every year. And remember who killed the random walk model says, as others say too, it's not our novel view. But most practitioners believe that the stock market is not perfectly elastic. That when you need to buy, buying of a trillion dollars of stocks if

somebody turns up and says I'm gonna buy a trillion dollars of stocks next year that should move stock prices by a pretty reasonable amount we would estimate it from our model we would estimate it that like it would move the market up by three or four percent there are other other people think it would be more than that some people would think it would be less than that most traditional financial economists would say it's not gonna move the market at all the market is fully elastic But that's ridiculous, I think, or at least I wouldn't call it ridiculous. I would say I would challenge that. I don't believe that. So, you know, we've had a trillion dollars a year of these buybacks, almost no IPO calendar. I mean, when we had a big IPO calendar, it was SPACs, but SPACs are not IPOs. SPACs were just money that was going to buy more stocks. It was like a kissing your sister thing. It wasn't like real equity issuance at all. It was just.

When that wash so so anyway, I think we're here because we've had these buybacks. We've had not much IPOs We've had businesses that have been capital light for a really long time and we have extrapolators They want to buy more stocks when they go up and we have static asset allocators that just don't want to sell their stocks unless they go up a lot and they and they need to rebalance and we have almost no fundamental value investors anymore so stocks us stocks going up and up and what's happening now maybe it's about to change. Okay, so Victor, the operating earnings of the S&P have been growing about 24%, talking about blended operating earnings, and they're expected to grow by 24% over the next 12 months. That is a very rosy situation. So the forward PE for the S&P 500 based on that is 20, which is higher than average, but not what you have seen at bubbles, which is like 30 or something like that. Why are you saying that the forward expectations for the S&P 500 are only 6% or in that modest thing? Like, why is it crazy that if the earnings are growing so much as they are now, the stock should go up and maybe the multiple should go up as well? Sure. So first of all, let's see.

So first of all, why is it, not first of all, but your question is, why is it that we at Elm and 20 other market observers, investment banks, investment managers, think that the long-term expected return for the US stock market is 6%, just given that we're seeing 20% plus earnings growth recently and into the future? What are they thinking? What are we all thinking? Why

don't we just believe that we're going to have 20% earnings growth for a long time and so stocks are going to deliver 15% returns? What are we missing in saying that stocks are only going to have a 6% return in the presence of tremendous earnings growth? Well, the answer is that when thinking about earnings, most people think about cyclically adjusted earnings because just as let me let's say that next year, We're in a let's just say that we're sitting at some point in time like we're we're we're in the end of 2008 and and and last year's operating earnings for the S&P 500 were like negative where it's right at the end of 2008 and or early 2009 and we're like, oh my god Operating earnings etc for these companies was negative last year and we expect next year Operating earnings are going to be negative again, or they're going to be close to zero Why why is anybody buying equities at all? I mean clearly equities are worth zero because next year's operating earnings are going to be close to zero like why would you buy any equities right and and so what people realized is that we don't want to base your valuation on equity of the equity market based on last year's earnings or next year's earnings you want to base it on some sort of cyclically adjusted realistic long-term earnings path. I mean, if we get 25% earnings growth of the US stock market for three or four years, then GDP is basically going to be 50% corporate earnings, where historically corporate earnings have been 8% of GDP or 6% or whatever, much, much lower. So the fact that we might have some strong earnings growth last year, next year or whatever, shouldn't make us think that earnings are on some moonshot trajectory where the whole economy is going to be corporate earnings. That just isn't going to happen. There's something called competition where competition erodes margins. And we won't go into like all the different ways that sometimes in earnings in earnings bull phases, earnings tend to get a little overestimated. Like we get a lot of a lot of today's earnings are markups of cross holdings, a fair amount.

as markups of cross holdings, et cetera. But we won't go into a deep dive on earnings. But in general, a pretty simple thing to do is to say, I'm going to use the last 10 years of earnings adjusted for inflation, adjusted for payouts. I'll use that as my base case. And that's what we do. And all of these other 19 investment houses that are saying, I think the returns are going to be 6% over the next 10 years. They're doing things differently than we're doing it. They're doing things differently than each other. But they're all basically

trying to think about where are earnings? I mean, ultimately everybody's like, where will earnings be 10 years from now? And what will the multiple be on those earnings 10 years from now? And they're coming up with a 6% return despite some tremendous earnings momentum that we've had in the past few years and that we expect to continue for a while too.

Right. But by forecasting 6% longer terms, you are making a call that the trend that has caused earnings to go up so much, which is five companies, the hyperscalers spending so much on building out data centers, NAI, which is flowing to mainly the semiconductor companies, the construction companies as well, that that is going to stop or is not sustainable. And I'm not like, do you agree that you are kind of making that bet? that leads me to say, like, what are your thoughts on this? Like, do you think this continues for 18 months, 18 days, 18 years? Like, I mean, 18 years seems unlikely. It's going to grow. I mean, I'd say approximately zero that it grows at this rate. But like Nvidia's revenues are still growing at 70 to 80% your year. And it's the biggest company in the world. Like just the earnings momentum is just so enormous.

It seems to me like if you say they're overvalued, we can catch up to fair value pretty quickly. If the stocks don't move and they're earnings just for 18 months, if the earnings grow as fast as they are now, the stocks will be cheap. Yeah. I mean, look, margins are very high. We're already starting to see competition within US companies, between US companies and with foreign companies as well. There's always competition. And as I say, historically, look at earnings in 2009, 2010, 2011, and you'll see even stronger earnings growth over those periods. And could we extrapolate that for how long? No, I mean, it kind of settled down and it got into a more normal pattern of growth that was mostly driven by retained earnings and share buybacks. And so I think that Look, I mean, we've had tremendous technological developments over the last 30 years, right? Go back to 1996. So from 1996 to today, I mean, think about it. We have just massive technological changes that were tremendously positive for productivity. We had the internet. We had fracking that just created this huge windfall.

to all kinds of energy companies. We've had consolidation in the financial industry, where the financial firms have been massively more profitable.

And recently, we've had just tremendous growth in the earnings of the MAG7, or whatever, with all different business models and tremendous margins. Well, go back over that whole 30-year period, earnings growth has been quite nice, but it hasn't been ridiculous. When once you account for share buybacks and retained earnings, earnings growth, I don't know. I mean, I'm just going to throw a number out there. I think that earnings growth adjusted for those things has been a couple of percent per year adjusted for buybacks and retained earnings. So buybacks is company takes its cash flow and buys back its stock in the market, which boosts the earnings per share because the shares outstanding goes down retained earnings is just the earnings that the company keeps. I understand why they can invest it and that can grow earnings too. So when you adjust for those things and just think about like core like core earnings growth from productivity or from great things happening to companies, it just hasn't been that great over perhaps the best.

Period of time that any that any economy any country any set of companies has ever experienced We're just at mega high margins on a bunch of these businesses and you can just see the competition It's coming from every direction. I mean everybody's competing with the video right now It's like oh you guys make some really nice chips. You mean you design some really nice chips Maybe we'll try to design some nice chips too and and so on I mean some of the motes are bigger than other motes, but there's There's no moat lasts forever. Victor, why? I understand the argument that when a company buys back its shares, it is boosting the argument that it's artificially boosting its earnings per share. But why is retained earnings? Why shouldn't that count as earnings? I'm a little confused. The retained earnings count as earnings. But when you look at earnings growth, you have to realize that the retained earnings are getting invested.

are being invested by the company in in in output or whatever and And that's growing earnings per share too. So a company who retain forget about buybacks for a second Just say a company that retains its earnings is going to grow faster than the same company that doesn't retain its earnings that pays everything out as dividends So the retained earnings count as earnings when they're earned. I'm not saying that I'm saying that the retained earnings are also boosting earnings growth when in a predictable manner

and we need to take that out when we're thinking about like real earnings growth just come by businesses getting better and better. So because if you had paid me the earnings out as a dividend instead of retaining them I would have invested them in the stock market or whatever and so that you need to adjust earnings growth when you're thinking about like how great has earnings growth been per share.

But you need to adjust for retained earnings and for buybacks That does make sense. Okay. So yeah, like in the 1950 you'd have a steel company It would pay out a lion's share of its income as dividends It wouldn't have that money to invest in its business which would cause earnings to grow or To buy back its own stock which would not cause net income to grow but it would cause earnings per share to grow because that Okay that That makes sense. And we wrote about, we wrote about that a while ago when we started to use what we call the PKAPE measure for CAPE instead of CAPE PKAPE, which adjusts for payouts effectively or lack of payouts sometimes in earnings growth. Okay. So yeah, Victor, I think the fate of the S&P 500 over the next two years, it really does depend on is the spending on AI CapEx going to continue and what is the return on that?

Cap X and if the return is bad and the spending goes down, I think earnings are going to be negative, perhaps massively negative, at least if you include like potential markdowns and the depreciation that's kind of baked in. So I feel like Victor, if I was you and I had a brain that was as big as you, I would be spending like 80% of my time focusing on that question rather than finance questions that do matter, but are an indirect kind of output. rather than an indirect input, a direct input into this. So like what is your thoughts on this? And if your thought is, Oh, I'm not spending that much time thinking about it. I'm thinking about these other things. Why are you thinking about these other things? Well, great question. Well, first of all, we at Elm, me, my partner and me and my partners, like we don't really think that our brains are too big and that we can really figure this stuff out. So we kind of feel like we're leaving it to many, many experts in the market to think about what's happening with earnings, to price these different companies relative to each other correctly. And two other points, but at Elm, we're like, okay, what we want to do is give you a really low cost investment strategy that we think makes sense. And again, we had a team of

people trying to figure out what was going to happen with S&P 500 earnings better than everybody else.

We'd have to spend a lot of money on that and charge our clients high fees instead of the 12 basis points. We charge them the third thing and maybe the most important thing perhaps is that let's go back to who killed the random walk that we started with and I would say that I guess our belief is that S&P 500 earnings kind of don't the next the next two years. Stock market performance is mostly going to be determined by. the behavior of extrapolators interacting with corporate equity activities. So if our belief is that if we see a lot of issuance, which it seems like we might be seeing IPO issuance, secondary issuance, sales from insiders of stock ownership like in SpaceX, a lot of people are going to be selling SpaceX.

And they're not going to invest all the money in the stock market because they have to keep some of it to pay capital gains taxes. And finally, a reduction in buybacks of the big buyback companies, you put those things together. And if that's going to make a delta of 1 to 2 trillion, we think that's going to weigh on the market. And if the market gets weighed upon, eventually the extrapolators are going to be like, oh, the gig's up, the party's over. and we'll see markets go down. So that's like a scenario. And what S&P 500 earnings are is totally immaterial to that. But at the end of the day, I think that with the overall market level is not being determined that much by fundamental investors anymore, because as I said at the beginning, the predominance of asset allocators, the predominance of people that are deciding how much to have in stocks, How much to have in bonds are either static investors, passive asset allocators, or they're extrapolators. And the asset allocators that are thinking about the expected return are like in a pretty small minority these days. Yes, there's Berkshire Hathaway. Yes, there's other investors. Yes, there's GMO. But there's not a lot. And most of those fundamental asset allocators are like mostly out of the market because they're seeing 6% returns compared to a 5% long bond. And they're like, eh, I don't need to own much equities right now, or I need to own as little as I can get away with. So they've already done their thing. And so I kind of feel that that's where we've spent our time is trying to understand the dynamics rather than it just doesn't feel like earnings.

The next year of earnings is that material I mean it certainly gives this certainly gives this story to what's going on right so it's like oh the stock markets up because earnings have been so good but it's kind of like maybe the stock markets up because the extrapolators are running the show plus companies are buying back their stock maybe that's the whole story. And the rest of the stuff it's a little bit. It reminds me, have you read this short essay by Richard Sutton called the, oh, what's it called? I'll come back to that in a second. I'm drawing a blank. It's like called the bitter lesson, the essay called the bitter lesson, where he's like, it's a really fun short article by one of the fathers of AI and machine learning. And he says, it's kind of sad that All of us machine learning guys, we were trying to build these smart AI models by trying to imbue them with human thinking. And it turned out that all that we needed to do was just throw a ton of computation. We needed a few good insights. But just the more computation we threw at the problem, the better the AIs did. And we didn't need to model the way that humans think. And that's what he called the bitter lesson. And it might be that the bitter lesson in finance is that you don't need to think that much about earnings and stuff like that and fundamentals because the market isn't driven by fundamentals. It's just driven by these static asset allocators, these extrapolators and corporate supply and demand and other shocks and other supply demand shocks to the system. I don't know. Okay. Finally, we have our intellectual cage match. I, so you think earnings don't matter. Like I disagree. I think you look at the stocks that are up the most.

They're the stocks whose earnings have crushed the most and whose forward expectations of earnings have risen the most. Those are all in the semiconductor space. To me, that seems like a little bit rational, and it makes sense. Why am I wrong? You're not wrong, but we're talking about two different things. I'm talking about the stock market and aggregate, and I'm also exaggerating. I'm not saying that earnings don't matter. Earnings matter a lot to us as investors. I really care about earnings. our asset allocation depends on earnings. So I care about them a lot. I'm saying the market dynamics of the market in aggregate, I don't think is as affected by earnings. I don't think the earnings is like the central thing. I would stand by that. Not that it doesn't matter. I agree with you that in the cross-section, earnings matter a ton. The earnings of NVIDIA and earnings growth, again, I agree with you that that more or less the cross section of stock valuations

is like reasonable and earnings matter a lot and the cross section of stock prices is like being driven by fundamental investors, stock pickers who really dig into stocks a lot and they come up with reasonably good prices except every once in a while for five or 10 or 15 different stocks where some people are going bananas and then the fundamental investors like kind of get out of the way and say what will be will be and they certainly don't go short them but overall in the cross section I would agree with you that earnings matter a lot and they seem to be reflected earnings and earnings growth expectations seem to be reflected in individual companies fairly well. So earnings and earnings expectations matter a lot to individual companies and perhaps sectors but to the overall broad market you think They don't matter maybe as much as they used to or as much as people think. So if, yes. Perfect. That's what I think. So to the S and P 500, the stock market, the Vanguard total world market, the MSCI world in it, if earnings is not a primordial primary driver, what are the top drivers? Extrapolators, static asset allocators and supply demand changes, changes to shares outstanding basically issuance IPOs secondaries stock buybacks and and there are other shocks on the demand side, right? Like how much money are our white collar invest? How much money are relatively affluent people saving in their 401ks and then gets rooted into the stock market? So it's not all the corporate activities, but there's also flows. There are also these savings Changes in savings flows a year on year that are also important. So if we see we've seen a lot of Net contributions net contributions to 401k and IRAs in the US and that money's gone into the stock market It looks the same as stock buybacks in terms of impact And that changes or slows down or whatever then that's another net change to this number amount of equities and it has a lot of price impact, we think, that markets are not that elastic with respect to changes in supply and demand of equities. Do you think that those flows can be forecasted and therefore you can forecast proper asset values based on those flows? Because to me, it seems very hard being like, oh, in Japan, they have a bunch of old people, so they're going to buy less stocks, whereas in the US, We have a stock, like it seems very hard to like have that be something that could generate a kind of a sharp ratio. But I wonder if you can, I think it's, I think it's hard. I think in general, the best estimate is kind of what happened last year. And then every once in a while, you might see something that seems to be changing. So I think with respect to individuals, I think it's, it's

hard. It's hard. I think with respect to corporations, it seems a bit easier because we hear about their plans and they're like, Hey, We have a stock buyback program. We're going to buy back 100 billion of our stock next year. And they tell you that. And then they start doing it. Or they say, we're not going to buy back 100. So I think that different components have relative difficulties of predicting. I think this investor side, the demand side, I think is in general much harder than the corporate side.

Corporate buybacks, let's say there's like three stocks in the S&P that are doing a ton of buybacks and 497 of them are not doing any buybacks at all. Do the buybacks boost only the three stocks that are doing the buybacks or does it boost the market in general? Good question. So let's see. So I think the first order thing is it's sort of boosting the market in general. The the index funds the index funds are going to be kind of passive with respect to this flow So a company comes to buy back at shares and now the number of shares that that company is going to have outstanding go down and the index fund is going to say, okay, I am I'll sell you those shares I mean they might not meet at the exact same moment in time But let's say they do they're like I'll I'll sell you those shares because I need to have less shares of you in my index And then they turn around and what do they do with that cash? They have to buy all the other companies. So if everybody were an index fund, I would say that the buybacks actually could have the impact of counterintuitively of raising all the prices of all the other companies and leaving the price of the company buying back at shares unchanged. Maybe I've got that wrong because it does seem pretty counterintuitive, but I think that might be how it goes.

I think that might be how it goes. But the overall effect is that it's lifting the whole market. It's lifting the whole market. But I think the mechanism could well be, I'm sorry, again, it depends. Because then, so that's the index fund. So if everybody were an index fund, I would say, oh, it's raising all the other stock prices, which is weird. But I think it's correct. Or maybe it's not, but I think it's correct. But then let's put the index funds aside. And now we have everybody else, just regular people. And now whoever owns this company that's buying back at stock, likes this company. They chose it. They're a stock picker. They bought that company. Company's buying back at stock. And they're like, I don't want to sell you my stock. I have a capital

gain. I don't want to pay capital gains tax. Or I just love this stock. I don't want to sell it. And that pushes that stock price up. And eventually, they're like, OK, fine. I'll sell it to you.

the buyback is pushing up the stock of the company that's buying back at stock with respect to people who are not index funds, people who are like care about the valuations and so on. But then when they do get that money, when they do eventually sell that stock, they buy some other stocks, and that pushes the other stocks up. But the big impact, but that doesn't have much impact on all the rest of the stocks, there needed to be quite a lot of impact on that stock to get that marginal person to say, okay, fine. I'll sell it back to you, Mr. Company, and I'll pay my capital gains tax or I'll find something else to invest in, fine. You've pushed the price up far enough where I accept that. And of course, there's a lot of closet indexing, which is more like the index funds where they do that. So net, net, net, I think the whole market is pushed up. And depending on who's depending on how big the buyback is, Sorry, depending on the relative weights of kind of indexers and closet indexers versus fundamental guys, is going to determine what that shock is. But I would say the company that's buying back shares, that stock price goes up. Everybody else goes up. The whole market goes up. That makes sense. Victor, remind us what extrapolators do and why are they, as you say, the critical ingredient?

in your model. So within our model, we have different investor types that come up with their expected returns for stocks in different ways. And then when they have their expected returns, that drives how much they want to own of stocks. So extrapolators are people who estimate the future return of the stock market based on its past return. They're extrapolating the past return into the future. We model them as doing that when it dampened way. We don't say whatever the return was last year over the last five years is their expectation for the future. But when it's high, we make that expectation higher. When it's low in the past, we make it lower. We use it. We dampen it a little bit or a lot. And that's what they do.

And that's how they set their asset allocation. So they kind of create a bit of reinforcement for bull markets and a reinforcement for bear markets. So Victor, I am someone, I look at the extraordinary earnings growth in the

semiconductor space, the extraordinary stock performance. I do think it's going to continue. So yeah, I just, I will go in the record in late July, July 21st, like I'm, I remain bullish in semiconductors, I think they're going to do. So basically I'm an extrapolator. And extrapolators can be right and they can be wrong. Why on average have you found that extrapolators tend to have sub-market performance? And how does extrapolators different from momentum people? Because it sounds like momentum is such a positive way of putting my view of semiconductors or positive momentum, whereas extrapolators is such a less positive view to put it. And what's the difference between extrapolators and momentum people? And if you were to...

To determine Jack you're an extrapolator or Jack you're a momentum person. What questions would you have to ask? Sure, I think this question of What's that that an extrapolator or return chaser? Gosh, it just sounds like the same thing is a momentum investor. What the heck? How could it be that return chasing or being an extrapolator is bad? But being a momentum investor is good. They sound like exactly the same thing. Don't they? I mean, what the heck? And we think this is a fantastic question and puzzle, something that we've thought about and written about for more than 10 years. And so I would just say, look, we have a hypothesis. We don't have an answer. It's something we've thought about and written about a lot. But I just want to start off by saying, I'm not really sure. Our working hypothesis is that the difference between what does a momentum strategy look like? A momentum strategy is that when Momentum is positive when today's price is higher than last year's price or the moving average price when momentum is positive I I overweight my position by X and when it's negative I underweight my position by X or by Y or whatever And that can apply to prices. It also can apply to fundamentals like earnings. Go ahead. Sorry Well, you're not trading earnings. So you could say that but but that's not that's not a price that you know, like applying it to earnings is fine, but you have to look for what's the action you're taking in the marketplace if you want to test it as an investment strategy. You get data points four times a year instead of 250 times a year. How many trading days are you sure? But I'm saying you could follow a strategy that's based on earnings momentum. And that's something we haven't looked at. And again, let's just forget I said that. Sorry. I don't want to do values. Go ahead. And also there's like the Again, that sometimes you're talking about the cross section,

you're talking about individual stocks, we're talking about the market. But anyway, so let's go back. So momentum means that when momentum strategy is implemented by going overweight or underweight, a fixed amount of exposure when momentum is positive or negative, respectively. And so you could really think of it as this binary operation, this binary trading strategy. So I'm sitting there, I come, I look at some asset.

Momentum goes positive. I normally am 50% exposed to the stock market. I'm a momentum investor. Momentum's positive. I go to 75%. As soon as momentum's negative, I go to 25% exposure. And I stay there until momentum goes positive. And then I go back to 75. And that's a momentum strategy. That's well-defined. Everybody agrees what is a momentum trading strategy. There's literature. There's everything. Everybody knows what a momentum strategy is. There's nothing, nobody has defined, sorry, there's not agreement on what is a return chasing strategy. There's not really any agreement on that. There's not a consensus. Partly there's not a consensus because when people are testing for strategies, they test strategies that have done well historically and momentum's done well. So there's a whole literature on what is it that we're testing, it's this momentum strategy and that's what it looks like. Return chasing is kind of up to each researcher a little bit to define what it is that we mean by return chasing. So when I say return chasing, we have defined it very specifically, we've defined it within our multi-agent model who killed the random walk, we've defined what the return tracing strategy is. And that return tracing strategy is when returns have been good historically, the better that returns have been historically, the more we expect returns to be good in the future and the more equities we want to have. So you can see that the big difference between return chasing and momentum is that return chasing is not a binary.

strategy as so imagine stocks are going along a little bit and They're doing well. So now all of a sudden Momentum has gone positive. So I say, okay, I'm gonna be 75% in equities because momentum is positive That's the momentum strategy the return tracing strategy is like oh stocks have been doing well the last year or so whatever I'm more bullish on the future. I'm gonna go from 50 maybe to 55% in equities Then if the market goes up some more and the returns historically have been even better, I'm going to

say, oh, they've been even better. My future expectation is even better. I'm going to go to 60% in equities. And eventually, when returns for the last year have been 20%, I'm going to say, oh, that's wonderful. I'm going to be at 75% in equities. So it's kind of this, the way that we're defining it is the slow moving pattern.

of the extrapolator getting more and more invested in the market, then the market starts going down or whatever, and they're kind of slow coming out. So the way that we've modeled it is that. That's how we've defined what a return chaser is or an extrapolator is. Now other people have thrown out other ideas for it. So I think Cliff Asnes at AQR has said, a return chaser is somebody who's doing a momentum strategy, but with a five year look back. Okay. That's also testable, et cetera. We've kind of looked at that within our model. We've looked at it with historical data. It's not as good an explanation in our opinion as what I just described. But that would be another way of defining a return chaser. That's what I was thinking while you were talking of a momentum person.

tracks what happened last day, last week, last month, a return chaser is someone who looks at the last year, the last three years, the last five years. And it sounds like that's kind of what you're saying. But you're also saying that. Yeah, not really. Yeah, we don't think it's yeah, we don't think it's as much a horizon thing as it's this binary thing. So it's the momentum is binary. And go ahead. Sorry. And return chasing is this more continuous thing. We think that's more at the heart of it. But there's a lot of debate. And I don't want to say that we're right and anybody else is wrong. That's how we look at it. And then so you said, how can I tell if you're a return chaser, what questions would I have to ask you? And I guess what I would say, given what I just described, is I would say to you, do you kind of take your dynamic asset allocation as like a binary chunk? Like when you get bullish, you go from 50 to 75 and stay there, or are you a little bit more slow moving? You're like, ah, this feels good. I'm going to get a little overweight. Oh, markets have been even better. I'm going to get more overweight. Oh, markets have been even better. I'm going to get more overweight. And vice versa, if you're on this more continuous move as a function of past returns, we would say you're a return chaser. If you're more of a binary chunky, I'll

give you this chunky thing, do this chunky thing, then we would say you're a momentum investor.

What we find is that historically and in our model, that the way we've modeled return chasing and the way we've modeled momentum, momentum does really well. Return chasing does poorly. That's a good question. I would say that I don't weight stock prices that high. I say I look more at earnings, forward earnings, and if I think the forward earnings are too high or too low, To me, at least the earnings that I have on my fiscal AI terminal, I don't have a Bloomberg terminal, but at least what's available to me, like the earnings estimates and the revenue estimates for lamb research are just too low. And if lamb research only earns how much the analysts expect, even the high estimates, then memory prices are going to be as high as they are now. They're not going to come down, which means that micron is trading at eight times earnings at its.

I'm not going to change. So that's just a little bit more. But I'm more curious about you. You are really a fundamental investor. That's what you are. You're a value investor. You're a fundamental investor. You're thinking about cash flows. You're trying to estimate cash flows. You have a model for how earnings work. You're a cash flow fundamental investor. I'm talking about the market and aggregate. And when I talk about momentum and return chasers, they don't care about anything except for price. They don't even know what earnings, they don't even know what any of, they don't even know what any, I don't know how to spell earnings, yeah, yeah. They don't know how to spell earnings. They don't know how to spell fundamental. They're just looking at last year's price and it turns out that's pretty good if you're a momentum investor and it's pretty bad if you, if the way that you incorporate returns is as a return chaser, it's pretty bad. And by the way, I just want to say that we haven't really talked about this, but within our framework, We also substantiate what Cliff Asnes and others have said, which is perhaps the best investment approach is this combination of value and momentum, where you're looking at long-term cash flows, you're a value investor, you're a long-term expected return investor, and you're also incorporating momentum, either as a risk signal or whatever. But when you put momentum and value together, I think as Asnes and others have said, that's the most robust approach to investing in

any market. And that's kind of what we've modeled our approach at Elm on is this combination of value and momentum, trying to get like the best of passive and active put together. And we were really influenced by this very seminal paper by Cliff and his co-authors, co-researchers called Value and Momentum Everywhere. I think for me, that was probably the biggest thing that changed my mind about markets that I read in that I've ever read as a single thing. And it was fantastic. Great, great paper. So in addition to your paper, who killed the random walk will also attach that cliff asses paper. Victor, why is it that having a static or having a binary zero or one, if momentum is of a certain quality and vigor, you increase your allocation. If you don't, you don't.

Why is it that that works so much better than having a modeled gradient thing? To me, it's not obvious that that would work at all, but you're clearly you've done the work and it is true. Why is that the case? This is so much fun, this conversation. You're like just asking these great questions, which I think are really like kind of hard questions without being able to go back and sort of have a little bit of a model in mind of things. So, So why is it that one does well and one does poorly? Well, as I said really early on, I think that it just depends on the relative capital of the two investor groups. So if there's a lot of capital that is doing this return chasing, then their activity is going to impact prices.

Momentum guys are like kind of price takers and not impacting prices at all You could really just think of these momentum investors as as front-running. I mean not in a legal sense, but getting ahead of the the Expected behavior of the return chasers so in a world in which These extrapolators are impacting prices because of their activity that we've described as such It's going to be a world in which momentum investing does well because the return chasers are impacting prices in a certain way There's still a lot of risk. I mean, it's not like the momentum guys are getting a sharp ratio of two or anything like that They're just getting a little bit better sharp ratio, but it's nice. It's really really nice and Because there's still a lot of uncertainty and a lot of shocks in the system But that's what's happening is that you could think of it as the return chasers are kind of impacting prices and the momentum guys are getting a little bit ahead of that and benefiting from these expected price changes that they're going to see over time

because of the somewhat predictable behavior of the return chasers. If we flipped everything around or if we made the momentum investors just as big as the return chasers in our model, we find that momentum no longer is doing anything great. Momentum loses everything in a world in which the capital of the momentum investors is as big as or not even as big as it doesn't even have to be as big as it just has to be really relatively big bigger bigger compared because remember these momentum guys have a lot of price impact because when they move they just do a lot right so at some point if momentum guys are big enough and if they really are synchronized they're going to make the markets really unstable but the momentum guys are not that big And that's because momentum just seems like kind of in some ways like kind of a bad investment approach. You say momentum's not that big, but isn't there so much capital that is explicitly in momentum funds directly or retail funds who are doing momentum indirectly that maybe they don't know it, but they are, and then hedge funds that are very aware of the momentum factor.

And technically they're neutral, all the factors, all the time. But like if there's any factor that they are long, it's, it's momentum, let's be honest. And they're very aware of it. Like, isn't there so much capital in momentum? I think not relative to the size of the markets. I don't know. Like how much, how much are the different hedge funds doing time series momentum and how much of that time series momentum is dedicated to equity market, to the overall equity market in particular. And of course, like a lot of there's momentum everywhere. and that there's so much stuff trade. You can have momentum in individual stocks, the cross-section of stocks, that's like one thing. You can have momentum in everything, commodities, currencies, interest rates, equities, everything that moves, crypto. I actually do understand the point of momentum as it goes very quickly from 50% to 70% or it's a big discrete jump, whereas the...

extrapolators return chasers are much more gradual in changing. So it is leading to the same thing of having a longer time horizon. They're just less quick to react there. So that, that does make sense to me. I, it doesn't make sense. Okay. So Victor, tell me about how you think it is appropriate to allocate capital. Tell, tell me about the dynamic index investing you do and what conclusions you drawing for your clients in terms of allocation to

foreign equities to relative to the US. Okay. So first thing is what do I want to invest in? And I really started thinking about this pretty late in my career. I think it was like 2000, 2001 post LTCM when I first started to really think diligently about how to invest. And my first reaction was to sort of do what a lot of my friends and people I respected were doing, which was kind of being this little mini David Swenson Yale endowment model at home in my home office. And I did that for a while and was like, this is no good. This is not financial freedom. This is super tax inefficient, et cetera. And then I kind of went back to the building blocks and said, what do I want to invest in? And I guess I didn't really think of it this way at the time. But since then, I've been thinking.

When it comes to risky assets, I want to invest some of my wealth in risky assets, and I want to have some, and maybe what I don't invest in risky assets, I want to have in safe assets. So that's the idea of how to allocate my personal capital, or I think how anybody should allocate their savings. So there's a million things to invest in. the infinite number of combinations of things that I could invest in, how do I narrow down what I want to invest in? And I kind of wound up with something that I call, or that we call at Elm, a five-star screen. Any risky asset that I want to invest in has to meet all five elements of a five-star screen. And again, I wasn't thinking about this explicitly at the time, but kind of implicitly, I think I was. And so the first thing that I want for a risky asset is I want a risky asset that where there's a good story as to why it should have a risk premium. Why investors should expect to earn a risk premium from the asset. And in general, that asset has to have this systematic, non-diversifiable, nasty kind of risk. The kind of risk where you lose money at the same time that you tend to lose your job and your home price goes down and everything sucks. So it should be some non-diversifiable.

non-diversifiable systematic risk that that asset is carrying. So that's the first thing. Well, lots of asset classes give you that stocks give you that private equity gives you that commodities like oil probably give you that 30 year treasury bonds probably give you that in terms of this inflation risk they bear. So that kind of gives me a bunch of assets. The next one is I want to be able to estimate that risk premium. I want to be able to verify, even though there's a good story there, I want to be able to verify and see the risk

premium. And this is where commodities drop away because it's so hard. How do we, the commodity don't, oil doesn't really have cash flows. It might be backward dated. It might be in contango, but who knows how to really think about that. So oil kind of drops out 30 year treasury bonds. It's hard to think about what's that risk premium in there because Whatever the risk premium is, it's small and hard to identify. But equities and real estate, great. Those things you can really think about. The risk premium is big enough that you can look at it prospectively and think about it based on earnings, et cetera. And so equities still stand up. The third screen is, I want to own this asset, but I don't want to have to take a lot of idiosyncratic risk. At the same time, I'm getting my systematic risk. I want pure systematic risk. So it's not enough.

that it has systematic risk, but that it should be pure. So that means like I don't want to just own NVIDIA because NVIDIA has systematic risk, it has beta, and it has idiosyncratic risk. So that moves me towards wanting this broadly diversified portfolios of things. The last two parts of the screen are that I want it to be very liquid and kind of transparent. And I wanted to have low fees in tax efficiency, or I forget how liquidity tax efficiency. and low fees and readily available. And so where that left me, it took me from billions of things I could invest in to wanting to basically be in low cost, broad market index funds and also maybe some real estate investment trust index funds as well. So that kind of got me to where I was going to be, what I was going to invest in. And then I thought, well, how much do I want to invest in each one of these asset classes? I don't want to just be, I mean, we've already talked about why I think that passive asset allocation just doesn't make sense. I think what does make sense is your portfolio should reflect the expected return of the things that you're investing in relative to safe assets, their risk and your personal degree of risk aversion. Also, this is known as the Merton share, which we wrote about in our missing billionaires book as a way of putting those things together.

And so I came to feel that I wanted to follow a dynamic asset allocation approach based on the long-term expected risk premium of these different asset classes and based on some sort of a risk metric when markets are really volatile and risky, I want to have less exposure and vice versa. Turns out that momentum is a good proxy for that when it comes to the stock

market. When momentum is negative tends to be high risk cases and vice versa. When the market's grinding up and momentum is positive, it tends to be in a lower risk state. And that's it. That's what we do at Elm. It's very low cost, fully transparent, rules-based. We're not doing any forward-looking analysis of different things. We're using a simple, transparent set of rules and charging really low fees to do it because it's all so easy.

It's it's it's nothing that we came up with ourselves. It's all based upon research of other people And so we don't think we should charge a premium for it. There's nothing proprietary about what we do and That's what we do at Elm we do it through separately managed accounts where we have I don't know about three billion dollars of client assets and separately managed accounts We have an ETF on the New York Stock Exchange with ticker Elm That has about 600 million in it. That's growing and is really nice It's been there for just a short time, but it's probably the only low-cost dynamic asset allocation ETF on the market. And we also manage some money for non-US investors too. And we've been doing it for 15 years and people tend to find us more than we tend to try to convince anybody that this is good. Most of our investors are people who...

We're doing it themselves and now pay us 12 basis points to do it for them rather than people that were paying Some private bank 2% to do it for them that come to us But that's the way it is kind of interesting that most of the interest is I think it's just who does what who does it resonate with what we're doing and that tends to be more do it yourself kind of investors and That's the whole story. We're based in Philadelphia And, uh, yeah, the team is growing. I love my colleagues and partners. And that's it. So you described the, the five qualities and it sounds like, so you want to understand why you're, you're being paid things. Obviously tax efficient is so important. And certainly there are so many topics in academic finance that matter less to long-term investment returns than tax efficiency. So I think that's really important.

But if I were to just open up your ETF, it seems like you're a little bit overweight, treasuries and cash and Europe and foreign markets relative to kind of the generic 6040 basket. Why is that the case? Is that because they're cheaper? So you're either along the value factor or you tell us more

about the strategy. Yeah, I forgot that was part of your question, too. So when we apply it to the markets today, what do things look like? So we look at US equities. Maybe in our ETF, the baseline weight for US equities is around 40%. But I think that we have less than 30% in US equities. Why is that? Well, as we've already stated, the long-term expected return of US equities relative to safe assets, in this case, tips is really low. It's around 1%. We don't want to have too many US equities based on that consideration.

On the other hand, US equities have positive momentum that's offsetting, but it doesn't fully offset. And so we're underweight. For non-US equities, expected returns look better than they do in the US. The earnings yield is higher. And so from a valuation point of view, we want to have more non-US equities relative to the baseline than we do in the US. And also, momentum is positive. We're in a low risk state. So that also makes us. So we're quite overweight non-US equities. And we're slightly underweight US equities. And when you put it, actually, I guess that's not quite right. We're quite underweight US equities. We're a little bit overweight non US equities. And when you put it together, we're net net a little bit underweight equities and a little bit overweight fixed income by about 10%. And within the fixed income complex at the moment, we're mostly in treasury bills, but we could move back more into tips and nominal bonds.

once interest rates stabilize and maybe if bond prices stabilize and kind of do a little bit better. Because right now we're kind of in this negative momentum state for tips and nominal bonds. So we're underweight them in our fixed income bucket. So that's kind of the whole picture of how the asset allocation is. It's not really dramatically different than the baseline a little bit. And if US equities outperform non-US equities, our investors are going to make less returns than they would. otherwise. And your underweight U.S. equities because the valuations are too high and the earnings growth is so far above the trend, you think it's going to mean revert? Yes, although again, we're not really specifically predicting earnings, but yes, I think that's fair to say. We're specifically predicting earnings by reference to the last 10 years of earnings adjusted for inflation and payouts and buybacks. So Yes, we are doing it. We just aren't doing it very much the way we're not doing it in any sort of detailed forward looking sense. We're saying, this is how we think about earnings. We think

about them with respect to past earnings. So yes, we do have really an explicit view about what earnings are going to look like in the future, but it's kind of based in this extremely simplistic model of the world. Yes. Would I be right in that?

cyclically adjusted earnings things is basically saying that the earnings over the past five, 10 years, the earnings growth has been too high. And that the market is expecting that is going to continue and then that's wrong. I wouldn't say that it's been too high. Historically, I would say that to the extent that people are owning equities because they think earnings are going to grow a lot over the next five years, We would say that we disagree with that. However, I would also say that we don't really think that's what's going on out there as we talked about earlier. We think that mostly what's going on is there's been buybacks. There's been extrapolators and that's why we are where we are. It's not so much. I don't think there are that many fundamental investors out there that are saying, oh, I'm overweight equities because earnings are going to be great. I think that's quite secondary in the whole story.

Yes, I mean to the extent that that that is our view. I agree with you. Yes So you think that a majority or a lot of capital is Invested in stocks because returns have been so good. I think so. Yeah Feels that way And also the static guys, right? I mean the static guys are just static, you know, they're just static. Yeah, yes, they don't They don't care about returns. They don't care about anything. They don't care about earnings, earning shmurnings, right? It's like, yeah, they're, they're static. They're just like, I don't care. I think the expected return is always the same. They, sorry. They're like, I think the expected risk premium and risk is always the same. And I'm just going to be 60 40 till, till the cows come home. Victor, your first book is called the missing billionaires. I haven't read it. I got to read it. I'll buy it after this. What's your second book about?

Well, we don't, we don't have a title, but it's a little bit of a prequel. And a lot of people, I mean, the missing billionaires, we wrote it for, for people that kind of work in the finance industry or people that really are deep, deep into finance and didn't have a book to read that was really interesting on personal finance. Like all the books that are out there on personal finance

were like too basic for them. And that's, I think that was our audience for the missing billionaires. And we were like, Totally shocked that we sold as many of that book as we did But we recognize that the missing billionaires is a bit of a challenging read But so so many people have said to us gosh I wish that you guys would write a book that I could give to my kids that I could give to my Young adult kids a more base like a prequel a little bit more basic that really talks about good personal financial decision-making But without going into all these equations and stuff that you had in the missing billionaires. And shorter, too, would be nice. Not 370 pages, but short, where you could almost read it in one sitting. And so that's what we're writing. We're writing a more basic, still every bit as kind of rigorous and as trying to bring the same rationale. And it still will be kind of interesting and challenging for any reader, we think, but in a good way. And we're really excited about it. It's coming along.

It's coming along. We have a working title, but we're not really sure. We're going to really work hard on getting a good title because I think The Missing Billionaires was a good title for our first book. And if we can come up with something that is appropriate and also memorable, that will be great. It is. Missing Billionaires is a good title. Missing Billionaires, part two, still missing. Yes. Victor, we'll leave it there. People can find you on X. Victor Higani at Elm Wealth. Thank you so much. Thank you so much, Jack. This was terrific. Really, I guess on the longer side, but I really love being able to... I just hope your audience will be as excited about your questions as I was in answering them. I think we really hit on some core, core investment things that are just...

hopefully going to be really useful to people as they think about them. We did. Thank you everyone for watching. Please leave a rating and review on for monetary matters on Apple podcasts and Spotify. Subscribe to the monetary matters YouTube channel until next time. Thanks. Bye bye.